assignment5 - electric field

This is a preview of the published version of the quiz

Started: Mar 22 at 7:09pm

Quiz Instructions

I willl make a tutorial in the dropbox. Hide the answers/video first

Question 1 10 pts

If the electric field exerts a 27 mN force on a point charge of -30 μC at a certain location in the laboratory, what are the magnitude and direction of the electric field at that location?
hint: mN means milli newtons so 10^-3)

- 900 N/C
- 810 N/C
- 900,000 N/C
- 9,000 N/C

Question 2 10 pts

A small object with a 5.0-μC charge is accelerating horizontally on a friction-free surface at 0.0050 m/s2 due only to an electric field. If the object has a mass of 2.0 g, what is the magnitude of the electric field?
hint: convert g to kg. uC to C. Remember Newton's second law F = ma

- 1.0 N/C
- 0.0020 N/C
- 4.0 N/C
- 0.0040 N/C
- 2.0 N/C

Question 3 10 pts

A small 0.050-kg insulating sphere carries a charge of -60 μC and is hanging by a vertical silk thread from a fixed point in the ceiling. An external uniform vertical electric field is now applied. If the applied electric field has a magnitude of 3000 N/C and is directed downward, what is the tension force that the silk thread exerts on the sphere?

hint: see the tutorial if you need help. Try first

- 0.52 N
- 0.19 N
- 0.71 N
- 0.31 N
- 0.41 N

Question 4 10 pts

What should the magnitude of a vertical electric field be so that the force that the electric field exerts on a plastic sphere of mass 2.1 g that has been charged to -3.0 nC balances the force that Earth exerts on it? ($k = 1/4\pi\varepsilon_0 = 9.0 \times 10^9$ N · m^2/C^2)

hint: convert to kg. weight (mg) is balanced by the electric force (qE)

- 2.1×10^6 N/C
- 6.9×10^6 N/C
- 7.8×10^5 N/C
- 1.5×10^6 N/C

Question 5 10 pts

A proton is placed in an electric field E = 800 N/C. What are the magnitude and direction of the acceleration of the proton due to this field? ($e = 1.60 \times 10^{-19}$ C, $m_{proton} = 1.67 \times 10^{-27}$ kg)

hint: F = ma

○ 76.6×10^{10} m/s^2 in the direction of the electric field

○ 76.6×10^{10} m/s^2 opposite to the direction of the electric field field

○ 7.66×10^{9} m/s^2 opposite to the direction of the electric field

○ 7.66×10^{10} m/s^2 in the direction of the electric field

○ 7.66×10^{10} m/s^2 opposite to the direction of the electric field field

Question 6 10 pts

A small glass bead has been charged to 1.9 nC. What is the electric field 2.0 cm from the center of the bead? ($k = 1/4\pi\varepsilon_0 = 8.99 \times 10^9$ N · m^2/C^2)

hint: nC is 10^-9 C. convert cm to m . Use the equation of the electric field. Its an inverse square law

○ 4.3×10^4 N/C

○ 8.5×10^{-7} N/C

○ 8.1×10^{-5} N/C

○ 8.5×10^2 N/C

Question 7 10 pts

Two tiny particles having charges +20.0 μC and -8.00 μC are separated by a distance of 20.0 cm. What are the magnitude and direction of the electric field midway between these two charges? ($k = 1/4\pi\varepsilon_0 = 9.0 \times 10^9$ N · m^2/C^2)

hint: see my tutorial. Try first.

○ 25.2×10^6 N/C directed towards the positive charge

○ 25.2×10^5 N/C directed towards the positive charge

○ 25.2×10^5 N/C directed towards the negative charge

○ 25.2×10^6 N/C directed towards the negative charge

○ 25.2×10^4 N/C directed towards the negative charge

Question 8 10 pts

The electric field at a point 2.8 cm from a small object points toward the object and is equal to 180,000 N/C. What is the object's charge q? ($k = 1/4\pi\varepsilon_0 = 8.99 \times 10^9$ N · m^2/C^2)

○ -16 nC

○ +16 nC

○ -17 nC

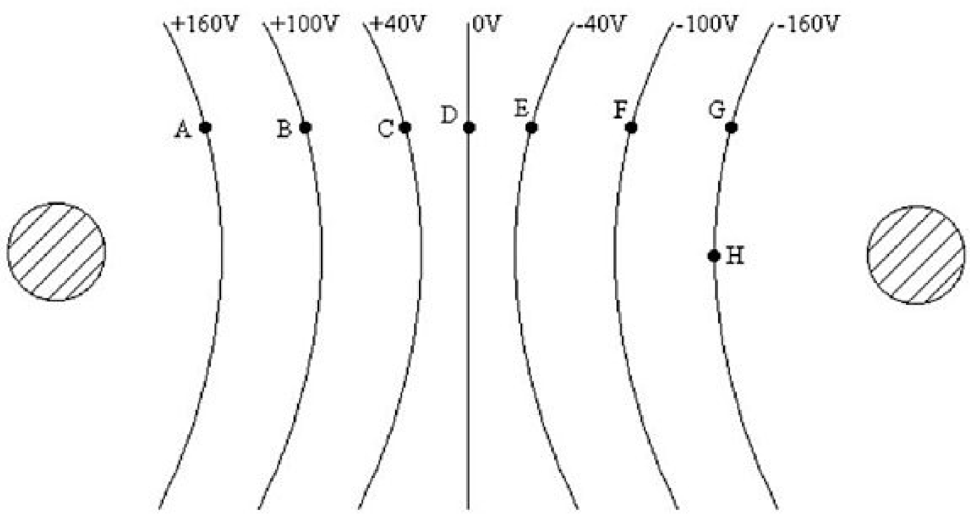
○ +17 nC

Question 9 10 pts

The equipotential surfaces for two point charges are shown in the figure, with the value of potential marked on the line for each surface.

What is the potential difference, $V_A - V_G$, between points A and G?

hint: it works like contour lines



- 0
- 160
- -160
- 320

Question 10 10 pts

A Xray machine uses a voltage of 50KV.

Each electron has an energy of

- 50KeV
- 50V
- 0
- 50KJ

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